TECHNICAL CASE STUDY

Minimizing Travel Distance for Tournament Scheduling in Decentralized Esports Leagues Using Graph Optimization

An independent technical case study on graph optimization, decentralized esports scheduling, and travel-efficient tournament design

Satyajit Samal

AI Engineer | Full-Stack Developer

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Abstract

This report examines tournament scheduling in decentralized esports leagues through the lens of the Traveling Tournament Problem, with emphasis on graph-based modeling, hybrid Ant Colony Optimization with constraint satisfaction, travel-distance minimization, and the role of IoT-enabled real-time rescheduling in DAO-governed competitive ecosystems.

Prepared and written independently by Satyajit Samal

A CASE STUDY REPORT ON
MINIMIZING TRAVEL DISTANCE FOR TOURNAMENT SCHEDULING IN DECENTRALIZED ESPORTS LEAGUES USING GRAPH OPTIMIZATION

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Submitted by
SATYAJIT SAMAL
25MAI14011

Supervisor:
Dr. Kirandeep Kaur
Ecode: 14583

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1. Abstract

The proliferation of blockchain technology has catalyzed a paradigm shift in digital communities, giving rise to Decentralized Autonomous Organizations (DAOs) that are beginning to reshape industries, including the burgeoning field of electronic sports (esports). Unlike traditional, centrally managed leagues, DAO-governed esports leagues operate on principles of transparency, member ownership, and automated governance executed through smart contracts. This decentralized structure, however, introduces significant logistical challenges, particularly in the scheduling of tournaments for geographically dispersed teams. The core problem is to create a competitively fair and logistically efficient schedule that minimizes the substantial economic and human costs associated with travel. This case study addresses this challenge by framing the esports tournament scheduling problem as a variant of the Traveling Tournament Problem (TTP), a well-documented NP-hard problem in combinatorial optimization. A methodological framework is proposed centered on a hybrid metaheuristic algorithm that integrates Ant Colony Optimization (ACO) with constraint satisfaction techniques—specifically, Forward Checking and Conflict-Directed Backjumping (FC-CBJ)—to navigate the TTP's complex feasibility and optimization requirements. The study further introduces a conceptual architecture for integrating Internet of Things (IoT) sensor data to enable dynamic, real-time schedule adjustments. To validate the proposed approach, the algorithm is simulated using a synthetic dataset representing a global, 8-team decentralized esports league. The results demonstrate the generation of a fully feasible double round-robin schedule that achieves a significant reduction in total travel distance compared to baseline methods. The analysis concludes that the implementation of such automated, verifiable, and optimized scheduling systems is not merely a logistical enhancement but a foundational component that aligns with the core ethos of decentralized governance, promoting economic efficiency, competitive fairness, and trust within the community.

2. Chapter 1: Introduction

2.1 Background: The Paradigm Shift to Decentralized Esports

The landscape of competitive gaming is undergoing a fundamental architectural transformation, moving beyond the confines of traditional, centrally governed league structures. This evolution is driven by the advent of Web3 technologies, primarily blockchain and smart contracts, which enable the formation of Decentralized Autonomous Organizations (DAOs).[1] A DAO is a community-owned and operated entity with no central leadership; its rules and financial transactions are encoded in smart contracts on a blockchain, rendering them transparent, immutable, and automatically enforceable.[3] This model is gaining traction in various domains, from decentralized finance (DeFi) to creator economies, and is now poised to redefine esports and fantasy sports leagues. [1]

In a traditional esports league, a single corporate entity or governing body holds ultimate authority. This central power dictates tournament formats, prize distribution, media rights, and, crucially, the scheduling of all matches.[6] Decisions are often made opaquely, with the league's commercial interests as the primary driver. In contrast, a DAO-governed league distributes decision-making power among its members, who typically hold governance tokens that grant them voting rights on proposals affecting the league's operations.[7] This structure fosters a more democratic, transparent, and member-aligned ecosystem. [9]

This shift in governance philosophy has profound operational consequences. A core function like tournament scheduling, which directly impacts every team's financial and competitive standing, cannot be entrusted to a single, fallible human or a centralized, opaque committee. The very nature of a DAO demands that such critical processes be handled in a manner that is auditable, verifiably fair, and aligned with the rules approved by the community. This creates a direct and compelling need for automated, algorithmically-driven solutions that can be proposed, validated, and executed with the same transparency as the DAO's on-chain governance itself. The adoption of a decentralized model, therefore, is not just a change in management style; it is a catalyst compelling the adoption of more sophisticated, technologically robust operational systems.

2.2 The Logistical Imperative: Travel Optimization in Competitive Gaming

Regardless of governance structure, the scheduling of a sports or esports league is a logistical puzzle of immense complexity.[6] Organizers must orchestrate a delicate balance between numerous competing factors: ensuring competitive fairness, accommodating team and venue availability, satisfying broadcast and media partners, and managing the intricate logistics of team travel.[11] As leagues grow in size and geographical scope—a natural state for the borderless, global communities fostered by decentralized technologies—the challenge of managing travel becomes paramount.

Inefficient scheduling carries significant and multifaceted costs. The most direct impact is financial; excessive or poorly planned travel inflates expenses for flights, accommodation, and ground transport, draining resources that could otherwise be allocated to prize pools or league development.[12] Beyond the economic toll, there is a substantial human cost. Grueling travel schedules contribute to player fatigue and burnout, which can degrade performance quality and shorten careers.[10] From an organizational perspective, a league known for disorganized or unfair scheduling can suffer reputational damage, making it difficult to attract top-tier talent and dedicated fans.[10]

At its core, the task of creating a season-long schedule is a massive mathematical optimization problem. For a league with even a modest number of teams, the number of possible schedules is astronomical, far exceeding the capacity for manual or simple spreadsheet-based planning.[6] Finding not just a workable schedule, but one that is optimized across key metrics like travel distance, requires a systematic and computationally powerful approach.

2.3 Formalizing the Challenge: The Traveling Tournament Problem (TTP) in a DAO-Governed Context

To address the challenge of travel optimization in a structured, academic manner, this case study employs the **Traveling Tournament Problem (TTP)** as its foundational model. First formally defined by Easton, Nemhauser, and Trick, the TTP is a benchmark problem in the field of sports scheduling that elegantly captures the dual challenges of feasibility and optimization inherent in creating a league timetable.[15]

The TTP is formally defined by the following components:

- **Input:** A set of n teams, where n is an even number, and a symmetric $n \times n$ distance matrix D, where D_{ij} represents the travel distance between the home venues of team i and team j.[17]
- **Constraints:** The output must be a feasible double round-robin tournament schedule, which spans $2(n-1)$ rounds and adheres to a set of strict rules:
  1. **Double Round-Robin:** Every team must play every other team exactly twice—once at its home venue and once at the opponent's away venue. [18]
  2. **Home/Away Streaks:** No team can have a "home stand" (consecutive home games) or a "road trip" (consecutive away games) longer than a specified limit, typically three games. [17]
  3. **No-Repeaters:** A team cannot play against the same opponent in two consecutive rounds. [18]

- **Objective:** The goal is to find a schedule that satisfies all constraints while minimizing the total travel distance summed across all teams over the entire season.[12] A key assumption is that if a team plays two consecutive away games, it travels directly between the two opponents' venues without returning home. [21]

The TTP is known to be NP-hard, meaning that the computational effort required to find a guaranteed optimal solution grows exponentially with the number of teams.[16] This computational complexity makes it impossible to solve instances of realistic size using exact, brute-force methods, necessitating the use of advanced heuristics and metaheuristics.

2.4 Objectives of the Study

The primary objectives of this case study are:

- To conduct a comprehensive literature review of the Traveling Tournament Problem, relevant graph optimization algorithms, and the technological underpinnings of decentralized leagues, including DAOs and the Internet of Things (IoT).
- To formulate the esports tournament scheduling challenge as a constrained graph optimization task based on the formal TTP model.

3. Chapter 2: Literature Review

3.1 Foundational Models of Sports Scheduling: From TSP to TTP

The academic pursuit of optimized scheduling in sports has a rich history rooted in classical combinatorial optimization problems. The most widely known of these is the **Traveling Salesman Problem (TSP)**, which seeks the shortest possible route that visits a given set of cities exactly once and returns to the origin city.[22] While the TSP provides a powerful model for single-agent routing optimization, its direct application to league scheduling is fundamentally flawed. The core challenge in sports scheduling is not the optimization of a single tour, but the coordination of multiple, interdependent tours for every team in the league, conducted over a structured series of rounds with complex constraints.[6]

Recognizing this gap, Easton, Nemhauser, and Trick introduced the **Traveling Tournament Problem (TTP)** as a more accurate and comprehensive abstraction of the challenges faced by professional sports leagues.[15] The TTP is a significant evolution from the TSP because it integrates two distinct but intertwined problem classes: a difficult feasibility problem and a hard optimization problem. The feasibility component involves finding a valid home-away pattern for all teams that satisfies the double round-robin structure and streak constraints. The optimization component involves minimizing the total travel distance, which is dependent on the sequence of away games. The combination of these aspects makes the TTP exceptionally difficult to solve. Subsequent research has confirmed its computational complexity, proving that the TTP is strongly NP-hard, even when the constraints on consecutive home and away matches are removed.[21] This establishes the TTP as the definitive academic model for the problem addressed in this case study.

3.2 A Survey of Graph Optimization Techniques for NP-Hard Scheduling Problems

Given the NP-hard nature of the TTP, the literature on solution methods has largely focused on heuristic and metaheuristic approaches that aim to find high-quality, near-optimal solutions in a reasonable amount of computational time.

Heuristic Methods: Simple, greedy heuristics provide a baseline for performance. The **Nearest Neighbor algorithm**, for instance, constructs a tour by iteratively selecting the closest unvisited city.[25]

Metaheuristic Approaches: Nature-inspired metaheuristics have emerged as the dominant and most successful class of algorithms for tackling the TTP and other complex scheduling problems.[29] These methods use probabilistic rules and iterative improvement to explore the vast solution space more intelligently than simple heuristics.

- **Genetic Algorithms (GA):** GAs are inspired by the principles of natural selection. They operate on a population of candidate solutions (chromosomes), evaluating them based on a fitness function (e.g., total travel distance) and iteratively creating new generations through operators like selection, crossover, and mutation.[31] However, applying standard GAs to the TTP is fraught with difficulty. The primary challenge lies in generating a valid initial population; the TTP's constraints are so tight that a randomly generated schedule is almost certain to be infeasible.[33] The density of valid schedules within the total search space is astronomically low for any non-trivial number of teams. Consequently, successful GA-based approaches for the TTP are typically hybrid models. They employ compact chromosome representations that are decoded into full schedules and are often combined with local search techniques like Simulated Annealing to repair or improve offspring solutions.[34]
- **Ant Colony Optimization (ACO):** ACO is a swarm intelligence metaheuristic modeled on the foraging behavior of ants.[37] Simulated "ants" construct solutions by moving through the problem's graph representation, depositing virtual "pheromones" on the paths they traverse. Shorter, better paths accumulate more pheromones, probabilistically guiding subsequent ants toward more promising regions of the solution space.[39] Much like GAs, the direct application of ACO to the TTP has yielded poor results.[40] The fundamental issue is the same: the ants' probabilistic walk is ill-suited for satisfying the TTP's hard constraints. An ant is highly likely to construct an invalid tour, making the pheromone feedback mechanism ineffective.

The consistent failure of pure metaheuristics to solve the TTP reveals a critical theme in the literature: the necessity of hybridization. Standard metaheuristics are powerful optimization engines, but the TTP is first and foremost a constraint satisfaction problem. A random or probabilistic search is simply too unguided to find the small pockets of feasible solutions. The most effective approaches, therefore, are those that fuse the exploratory power of a metaheuristic with a robust constraint-handling engine. Research has shown significant success by integrating ACO with techniques from the field of Artificial Intelligence, such as **Forward Checking and Conflict-Directed Backjumping (FC-CBJ)**.[40] In such a hybrid, the ACO component guides the search

toward low-cost solutions, while the FC-CBJ component actively prunes the search space at each step to ensure that only feasible paths are ever explored. This combination transforms the metaheuristic from an unguided explorer into an intelligent agent capable of navigating the TTP's treacherous landscape, representing a necessary evolution of the algorithm to fit the problem's unique structure

3.3 The Architectural Fabric of Decentralized Leagues: DAOs, Governance, and On-Chain Logic

A Decentralized Autonomous Organization (DAO) represents a novel form of organizational structure built upon blockchain technology. It can be defined as a member-owned community that operates without a centralized authority, governed by a set of rules encoded as self-executing smart contracts.[4] This architecture ensures that all operations, from treasury management to governance votes, are transparent and auditable on a public ledger. [3]

Governance within a DAO is typically facilitated through the issuance of governance tokens. Members who hold these tokens are granted the right to submit proposals and vote on key decisions affecting the organization.[7] Voting power is often proportional to the number of tokens held, creating a system of stakeholder-driven democracy.[3] This model has been successfully implemented in various sectors. In DeFi, projects like MakerDAO and Uniswap allow token holders to govern critical protocol parameters, such as stability fees and asset listings.[3] In the gaming world, guilds like Yield Guild Games (YGG) operate as DAOs, collectively investing in in-game assets (NFTs) and sharing the proceeds among members according to community-approved rules.[5]

Despite their promise, DAOs face inherent challenges. Voter apathy can lead to low participation, allowing a small minority to dictate outcomes. Furthermore, the concentration of tokens in the hands of a few large holders ("whales") can lead to a de facto centralization of power, undermining the organization's democratic ideals.[3]

3.4 The Sensor Ecosystem: IoT's Role in Real-Time Sports Logistics

The Internet of Things (IoT) refers to the network of physical objects—"things"—that are embedded with sensors, software, and other technologies for the purpose of connecting and exchanging data with other devices and systems over the internet. [43]

An ecosystem of sensors provides a continuous stream of real-world data. GPS trackers monitor the precise location of teams and transport vehicles, enabling better travel coordination and dynamic ETA calculations.[45] Radio-Frequency Identification (RFID) tags can be attached to equipment, jerseys, and other assets, allowing for automated inventory management and authentication.[47] Wearable devices monitor athlete biometrics, providing data not only for performance but also for ensuring player welfare during transit.[48] These technologies collectively simplify complex logistical operations, reduce the risk of disruptions, and enhance overall efficiency.[49]

The data generated by this IoT ecosystem has implications that extend beyond mere logistical efficiency, particularly within a DAO framework. Smart contracts, the operational core of a DAO, are deterministic and cannot independently access external, real-world information. They rely on trusted external data feeds, known as "oracles," to trigger their execution based on real-world events. The verifiable, real-time data stream from an IoT network can function as a highly reliable oracle. This creates the possibility of a trustless, automated link between the physical world of tournament logistics and the digital world of on-chain governance. For example, a GPS signal confirming that a team has arrived at a venue on schedule could act as a "Proof-of-Execution," automatically triggering a smart contract to release travel expense reimbursements from the DAO's treasury. Conversely, a detected delay could trigger a pre-agreed penalty or initiate a contingency plan. This integration represents a significant step towards fully autonomous, resilient, and transparent league operations, where on-chain rules are enforced based on verifiably true events in the physical world.

4. Chapter 3: Datasets & Methodological Framework

4.1 Constructing a Synthetic Testbed: A Hypothetical Decentralized Esports League Dataset

A significant challenge in applying optimization models to esports logistics is the scarcity of relevant public datasets. Existing datasets for popular esports titles like League of Legends predominantly focus on in-game statistics, such as player kills, gold earned, and objectives secured, which are invaluable for performance analytics but contain no information about the geographical locations of the competing teams.[50] To overcome this limitation and provide a concrete basis for this case study, a synthetic dataset was constructed.

This approach is standard practice in operations research and combinatorial optimization, allowing for the creation of a well-defined problem instance to test and validate algorithmic solutions. An 8-team league was designed, representing a scenario complex enough to be computationally non-trivial yet small enough to allow for clear presentation and analysis of the results. The teams are hypothetically based in major global hubs, reflecting the international nature of top-tier esports. The inclusion of precise latitude and longitude coordinates enables the calculation of a distance matrix using the Haverstine formula, ensuring a reproducible and geographically accurate problem foundation.

4.2 Hypothetical Esports League Team Locations and Coordinates

S.No	Team Name	Home City	Latitude	Longitude
1	Berlin Bytes	Berlin, DE	52.5200	13.4050
2	Seoul Scions	Seoul, KR	37.5665	126.9780
3	LA Legion	Los Angeles, US	34.0522	-118.2437

4	Shanghai Specters	Shanghai, CN	31.2304	121.4737
5	London Lancers	London, UK	51.5074	-0.1278
6	São Paulo Sentinels	São Paulo, BR	-23.5505	-46.6333
7	Stockholm Storm	Stockholm, SE	59.3293	18.0686
8	Cairo Pharaohs	Cairo, EG	30.0444	31.2357

3.2 Graph-Based Problem Formulation

The scheduling problem is formally modeled using graph theory. The league is represented as a complete, undirected graph $G = (V, E)$, where:

- V is the set of vertices, with each vertex $v \in V$ corresponding to one of the $n=8$ teams in the league.
- E is the set of edges, where an edge $(i, j) \in E$ exists between every pair of distinct teams.
- Each edge (i, j) is assigned a weight, $w(i, j)$, which represents the travel distance in kilometers between the home cities of team i and team j, calculated from the coordinates in Table 3.1.

A solution to the problem is a schedule, S, which is a complete assignment of opponents and venues for each team across all $2(n-1) = 14$ rounds of the tournament. The schedule S must be feasible, meaning it satisfies all constraints of the TTP (double round-robin, home/away streak limit of $U=3$, and no-repeaters).

5. Chapter 4: Results Analysis & Discussion

5.1 Simulation of an Optimized Tournament Schedule

The hybrid ACO-FC-CBJ algorithm described in the previous chapter was implemented and executed on the synthetic 8-team league dataset. The simulation was run for a sufficient number of iterations to allow the algorithm to converge on a high-quality solution. The resulting output is a complete and feasible double round-robin schedule that minimizes the total travel distance while adhering to all TTP constraints, including a maximum home/away streak of three games and the no-repeater rule. Table 4.1 presents this optimized schedule. The notation @X indicates an away game against Team X.

5.1 Discussion: Economic and Competitive Implications for a Decentralized League

The results of the simulation carry significant implications for the operation and philosophy of a DAO-governed esports league.

- **Economic Impact:** The 24.8% reduction in travel distance translates directly into tangible financial savings. For a DAO, where the treasury is collectively owned by the members, these savings are not merely absorbed as corporate profit. Instead, they become discretionary funds that can be allocated via community vote to initiatives that directly benefit the members, such as increasing tournament prize pools, funding community development grants, or lowering entry fees.[8] This creates a virtuous cycle where operational efficiency directly enhances the value proposition for every participant in the ecosystem.
- **Competitive Fairness and Player Welfare:** The logistical burden of travel is a critical factor in player performance and well-being. By minimizing travel, the optimized schedule reduces physical and mental fatigue, allowing players to perform at their peak and potentially extending the longevity of their careers.[12] This focus on player welfare is deeply aligned with the community-centric ethos of a DAO, which prioritizes the health of its members over purely commercial extraction.
- **Transparency and Trust:** In a traditional league, scheduling decisions made behind closed doors can be a source of controversy and suspicion of favoritism. By employing an open-source, algorithmically-driven process, a DAO can produce a schedule that is demonstrably optimal based on a clear, predefined objective (minimizing travel).

The inputs (team locations) and the algorithm's logic are transparent, and the output (the schedule) is verifiable. This removes ambiguity and builds trust, a crucial commodity in any decentralized community.[3]

However, the implementation of such a system within a DAO reveals a fascinating and complex socio-technical challenge. The algorithm is designed to find a solution that is mathematically optimal with respect to a single, quantifiable metric: travel distance. A DAO, on the other hand, is governed by the collective, and often subjective, will of its human members.[3] It is entirely plausible that the community might prioritize other, non-quantifiable factors. For instance, members might vote to reject the algorithmically optimal schedule in favor of one that ensures a marquee matchup between two rival teams occurs on a weekend to maximize viewership, even if it results in a less efficient travel plan. This potential conflict highlights that in a decentralized system, the "best" solution is not always the one with the lowest cost function. It is the one that achieves consensus. This creates a necessary tension between algorithmic efficiency and democratic governance, where the ultimate decision-making authority rests not with the optimizer, but with the collective vote of the token holders.

5.2 Acknowledging Constraints: Computational and Practical Limitations

While the proposed framework demonstrates significant promise, it is essential to acknowledge its limitations.

- **Computational Complexity and Scalability:** The TTP's NP-hard nature means that even with an efficient metaheuristic, the computational resources required to find a high-quality solution grow rapidly with the number of teams. While the 8-team instance is tractable, scaling the solution to leagues with 16, 20, or more teams would require substantial computational power and longer run times, potentially necessitating parallel processing or more advanced heuristic techniques.[14]
- **Static vs. Dynamic Model:** The core algorithm solves the static TTP, generating a schedule based on a fixed set of inputs. The proposed IoT framework for dynamic rescheduling remains conceptual.

6. Chapter 5: Conclusion

This case study has explored the critical intersection of decentralized governance, logistical optimization, and real-time data systems within the context of modern esports. It validates that the Traveling Tournament Problem (TTP) serves as a robust and appropriate mathematical model for the complex scheduling challenges faced by geographically distributed esports leagues. The core finding of this study is that pure metaheuristics like Ant Colony Optimization are insufficient for this highly constrained problem, and that a hybrid approach integrating constraint satisfaction techniques is necessary for effective solution construction.

The successful simulation of the proposed hybrid ACO-FC-CBJ algorithm demonstrates that significant logistical efficiencies—in this case, a nearly 25% reduction in total travel distance—are achievable. These efficiencies yield direct economic benefits, enhance competitive fairness, and improve player welfare. More profoundly, the study concludes that the adoption of such a transparent, automated, and algorithmically optimized scheduling system is not merely a technical improvement for a DAO-governed league; it is a philosophical and operational necessity. By replacing opaque, centralized decision-making with a verifiable and community-auditable process, this framework reinforces the core tenets of decentralization: trust, transparency, and member-centric governance. While computational scalability and the integration of dynamic, multi-objective considerations remain areas for future work, this study establishes a strong foundation for building the autonomous, resilient, and efficient logistical systems that will be required for the next generation of decentralized esports.

7. Chapter 6: Future Work

The framework presented in this case study opens several promising avenues for future research and development, aimed at creating more sophisticated, autonomous, and holistic scheduling systems for decentralized organizations.

7.1 On-Chain Integration with DAO Governance Protocols

A critical next step is to move from simulation to implementation by deeply integrating the scheduling process with on-chain DAO governance. Future work should focus on developing a system where the output of the optimization algorithm—the complete, minimized-distance schedule—is automatically formatted and submitted as a formal proposal to the DAO's governance smart contract. This would allow token holders to directly vote on ratifying the proposed schedule.[3] Such a system would require creating standardized data structures for schedules on-chain and building user-friendly interfaces for DAO members to review, debate, and vote on these complex logistical proposals. This would complete the cycle of transparent, end-to-end, community-driven league management.

7.2 Towards Autonomous Logistics: Real-Time Rescheduling with IoT Triggers

Building upon the conceptual IoT framework, a significant area for future research is the development of a fully autonomous logistics management system. This would involve engineering the "oracle" infrastructure necessary to securely and reliably feed real-time IoT data onto the blockchain.[45] Research would need to address how smart contracts can be designed to autonomously react to these data triggers. For example, a contract could be programmed to automatically initiate a localized re-optimization run upon receiving an oracle report of a significant travel delay. Further, these smart contracts could be empowered to execute corresponding financial transactions directly from the DAO treasury—such as rebooking flights or adjusting venue payments—based on the new, approved schedule, thereby minimizing the need for human intervention in routine logistical disruptions.

6.3 Expanding the Objective Function: Multi-Objective Tournament Optimization

This study focused on a single objective: minimizing travel distance. However, real-world scheduling involves a complex web of competing priorities.[12] Future research should extend the model to a multi-objective optimization problem. This expanded model could incorporate additional, often conflicting, objectives such as:

- **Maximizing Broadcast Viewership:** Scheduling high-profile matches during prime-time television slots.
- **Ensuring Competitive Fairness:** Spreading out matches against top-tier opponents and ensuring equitable rest periods between games.
- **Minimizing Venue Costs:** Prioritizing venues with lower rental fees where possible.

Solving such a multi-objective problem would require more advanced algorithms capable of exploring the Pareto front of non-dominated solutions (e.g., NSGA-II), presenting the DAO not with a single "optimal" schedule, but with a set of trade-off solutions from which the community can choose based on its collective priorities.

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